5 Week 5: Historical Perspectives on Observational Studies

5.1 Why Observational Studies?

In an ideal world, every causal question would be answered with a randomized experiment. Randomization creates treated and control groups that are comparable by design, allowing us to attribute differences in outcomes to the treatment itself.

However, in many important scientific settings, randomization is impossible. Ethical constraints prevent us from randomly assigning harmful exposures such as smoking or pollution. Logistical constraints make it impossible to randomize large-scale policies such as monetary interventions or school district boundaries. In other cases, the events of interest occurred in the past, long before researchers could design an experiment.

Despite these limitations, scientists have long drawn credible causal conclusions from non-randomized data. The challenge is that, without randomization, treated and control units may differ systematically in ways that also affect the outcome. Observational studies must therefore be designed carefully to approximate the balance that randomization would have created.

5.2 Examples Across the Sciences

Epidemiology: Smoking and Lung Cancer. Before randomized trials were feasible, early epidemiologists such as Doll, Hill, and Cornfield constructed comparisons between smokers and non-smokers who were similar in age, gender, and occupation. Their arguments relied on ruling out alternative explanations, demonstrating dose–response relationships, and showing consistency across multiple datasets. This case help illustrate how careful design and reasoning can yield strong causal conclusions even without randomization.

Economics: Returns to Education. Card (1993) used institutional features—such as proximity to colleges—to construct comparisons between individuals who were similar except for their likelihood of attending college. These comparisons resemble matching: treated and control individuals are made comparable by design, even though treatment was not randomized.

Health Sciences: Hormone Replacement Therapy. The Women's Health Initiative (WHI) famously contradicted earlier observational findings about hormone therapy. One must stress that the issue was not that observational studies are inherently flawed, but that their design must explicitly address confounding. Poorly designed observational studies can mislead, but well-designed ones can be highly informative.

Digital Marketing: Ad Exposure and Conversions. Firms often cannot randomize ad exposure due to platform constraints. Analysts therefore construct matched samples of users with similar browsing histories, demographics, and engagement patterns. The goal is to approximate the balance that randomization would have created, allowing credible estimation of the causal effect of ad impressions.

5.3 Design as Central

Our central message is that an observational study is not a "failed experiment," but an attempt to reconstruct the balance that randomization would have produced. This perspective leads to three guiding principles:

1. **Separate design from analysis.** The design of the study—how treated and control units are made comparable—should be done without looking at outcomes.

2. **Construct comparable groups.** Matching, stratification, and trimming aim to create treated and control groups that are similar in their covariates.

3. **Use covariates to refine, not rescue.** Covariates cannot fix a fundamentally flawed design, but they can greatly improve the quality of comparisons.

Matching is therefore a design-stage tool: it prepares the data so that simple outcome comparisons are meaningful and credible.

5.4 Matching as a Tool for Design

5.4.1 Basic Setup

We observe data (Y_i, Z_i, X_i) for units $i = 1, \ldots, N$, where $Z_i \in \{0, 1\}$ is a non-randomized treatment indicator and $X_i \in \mathbb{R}^p$ are pre-treatment covariates. Because treatment is not randomized, the distribution of X may differ between treated and control groups. If these covariates also affect the outcome, naive comparisons of Y will be confounded.

The goal of matching is to construct a subset of the data—a matched sample—in which the distribution of covariates is similar across treated and control units. In this matched sample, comparisons of outcomes resemble those from a randomized experiment.

5.4.2 Types of Matching

Matching methods differ in how they define similarity between units:

Exact Matching. Treated and control units are matched only if they have identical covariates. This is often too restrictive in practice, especially when covariates are continuous.

Nearest-Neighbor Matching. Each treated unit is matched to the control unit whose covariates are closest in Euclidean or Mahalanobis distance.

Mahalanobis Distance Matching. Distances are scaled by the covariance matrix of the covariates, giving more weight to covariates with low variability.

****Propensity Score Matching.** Units are matched on the estimated probability of treatment $e(X) = \Pr(Z = 1 \mid X)$. This reduces a high-dimensional matching problem to a one-dimensional one.

Optimal Matching. Rather than matching greedily, optimal matching minimizes the total distance across all matched pairs or sets. This approach is especially powerful when the number of treated and control units differs substantially.

Matching is a *design* operation: it creates a new dataset in which treated and control units are more comparable. After matching, outcome analysis can be simple—often just a difference in means—because the key work has already been done at the design stage.

5.5 Approximate Balance and a Chebyshev-Type Bound

5.5.1 Balance in Means

Consider 1:1 matching without replacement. Each treated unit i with $Z_i = 1$ is paired with a unique control unit $j(i)$ with $Z_{j(i)} = 0$. For a scalar covariate X, define the matched means:

$$\bar{X}_T^M = \frac{1}{n_M} \sum_{i:Z_i=1,\, i\in M} X_i, \qquad \bar{X}_C^M = \frac{1}{n_M} \sum_{i:Z_i=1,\, i\in M} X_{j(i)},$$

where n_M is the number of matched pairs. The difference in matched means is

$$\Delta_M = \bar{X}_T^M - \bar{X}_C^M = \frac{1}{n_M} \sum_{i:Z_i=1,\, i\in M} (X_i - X_{j(i)}).$$

Define the pairwise differences $D_i = X_i - X_{j(i)}$. If matching is successful, each D_i is small. If we even have a uniform bound $|D_i| \le \delta$ for all pairs, then

$$|\Delta_M| \le \delta.$$

Thus, pairwise closeness in covariates implies closeness of matched means.

5.6 A Chebyshev-Type Bound

Now suppose we do not have a uniform bound on $|D_i|$, but we know something about the dispersion of the pair differences. Define

$$\mu_D = \frac{1}{n_M} \sum_i D_i = \Delta_X^{\mathcal{M}}, \qquad s_D^2 = \frac{1}{n_M} \sum_i (D_i - \mu_D)^2.$$

If we view the matched pairs as a finite population of differences $\{D_i\}$, Chebyshev's inequality gives, for any $t > 0$,

$$\frac{1}{n_M} \left|\{i : |D_i - \mu_D| \ge t\}\right| \le \frac{s_D^2}{t^2}.$$

Interpretation: if matching produces a small variance s_D^2 of pair differences, then only a small fraction of pairs can have large imbalances in X. In particular, if s_D^2 is small, then most D_i are close to $\mu_D = \Delta_X^{\mathcal{M}}$, and the overall difference in means is itself small in magnitude.

More concretely, suppose we target a tolerance $\epsilon > 0$ for the mean difference $|\Delta_X^{\mathcal{M}}|$. If matching is designed so that s_D^2 is small, then:

- the average imbalance $\Delta_X^{\mathcal{M}}$ is small (by construction of the match),
- and Chebyshev's inequality guarantees that only a small fraction of pairs can deviate substantially from this small average.

In higher dimensions, we can apply the same logic to linear functionals $a^\top X$ for vectors $a \in \mathbb{R}^p$, or to standardized covariates. Approximate pairwise balance in X implies approximate balance in any linear combination of X, echoing the balancing interpretation of regression from Week 1.

5.7 Covariate Balance Measures and Their Connection to Chebyshev Bounds

Matching aims to construct treated and control groups that are comparable on observed covariates. To evaluate whether a matched design has succeeded, we use *covariate balance measures*. These measures quantify how similar the distributions of covariates are across treatment groups in the matched sample. They play the same role that randomization checks play in experiments: they assess whether the design has produced comparable groups before any outcome analysis is performed.

5.7.1 Standardized Mean Differences (SMD)

For a scalar covariate X, the standardized mean difference is defined as

$$\mathrm{SMD}(X) = \frac{\bar{X}_T - \bar{X}_C}{s_X},$$

where $\bar{X}_T$ and $\bar{X}_C$ are the treated and control means, and s_X is a pooled standard deviation, typically

$$s_X = \sqrt{\frac{s_T^2 + s_C^2}{2}}.$$

The SMD is a *unitless* measure of imbalance. It expresses the difference in means in units of the typical variability of the covariate. Standardization allows us to compare balance across covariates measured on different scales (e.g., age, income, test scores).

Why standardize? Raw mean differences depend on the scale of the covariate. A difference of 5 units in income is trivial; a difference of 5 units in age is large. Standardization puts all covariates on a comparable scale, allowing us to assess balance across variables with different units.

Interpretation. The SMD has a direct substantive interpretation:

- $\mathrm{SMD} = 0$ indicates perfect mean balance.
- $|\mathrm{SMD}| < 0.1$ is widely regarded as excellent balance.
- $|\mathrm{SMD}| \approx 0.2$ indicates moderate imbalance.
- $|\mathrm{SMD}| > 0.25$ is often considered unacceptable.

These thresholds are conventions, but they are used across epidemiology, economics, and the health sciences.

Why SMD rather than hypothesis tests? Hypothesis tests (e.g., t-tests) are poor diagnostics for balance because their p-values depend on sample size. In large samples, even trivial differences become "statistically significant," while in small samples, large differences may appear insignificant. The SMD avoids this problem by measuring *substantive* imbalance rather than statistical significance.

SMD in matched samples. After matching, we compute

$$\mathrm{SMD}_M(X) = \frac{\bar{X}_T^M - \bar{X}_C^M}{s_X},$$

where the means are taken over the matched sample. A successful match should reduce SMDs substantially, ideally below 0.1 for all key covariates. In practice, balance tables report SMDs before and after matching, and "Love plots" display these values graphically.

Connection to Chebyshev. Recall that in matched pairs, $D_i = X_i - X_{j(i)}$ are the pairwise covariate differences. If these differences have small mean and variance, then the matched means must be close, and therefore the SMD must be small. Thus, the SMD is a summary statistic that reflects the underlying pairwise balance structure created by the matching design.

Use in practice. SMDs guide both the design and evaluation of matching:

1. **Before matching:** identify covariates with large SMDs; these drive confounding and inform the choice of distance metrics, calipers, or exact matching constraints.
2. **After matching:** verify that SMDs have been reduced; if not, redesign the match (e.g., change distance weights, impose calipers, or use optimal matching).
3. **Reporting:** balance tables and Love plots are standard tools for summarizing balance in applied work.

5.7.2 Variance Ratios

Variance ratios compare the dispersion of covariates:

$$\mathrm{VR}(X) = \frac{\mathrm{Var}(X_T)}{\mathrm{Var}(X_C)}.$$

Values close to 1 indicate similar spread across groups. Variance ratios are especially important when covariates are skewed or when matching is performed on propensity scores, where mean balance alone may be insufficient.

5.7.3 Distributional Balance

Beyond means and variances, one may examine:

- empirical CDF differences,
- quantile–quantile plots,
- Kolmogorov–Smirnov distances.

These measures assess whether the *entire distribution* of a covariate is similar across groups. They are useful when matching on continuous or highly skewed covariates, or when treatment effects may vary across the distribution of X.

Together, these balance measures provide a comprehensive assessment of whether matching has successfully approximated the balance that randomization would have created. They connect directly to the Chebyshev bounds developed earlier: small variance in pairwise differences implies that most pairs are well balanced, which in turn ensures small SMDs and small bias in matched estimators under mild smoothness assumptions on $g(X)$.

5.8 Implications for Treatment Effect Estimation

Let $\hat{\tau}^{\mathcal{M}}$ denote the difference-in-means estimator in the matched sample:

$$\hat{\tau}^{\mathcal{M}} = \frac{1}{n_M} \sum_{i:Z_i=1} Y_i - \frac{1}{n_M} \sum_{i:Z_i=1} Y_{j(i)}.$$

Under the structural model

$$Y_i = \tau Z_i + g(X_i) + \varepsilon_i,$$

the bias of $\hat{\tau}^{\mathcal{M}}$ is:

$$\text{Bias}(\hat{\tau}^{\mathcal{M}} \mid X) = \frac{1}{n_M} \sum_{i=1}^{n_M} \big(g(X_i) - g(X_{j(i)})\big) = \frac{1}{n_M} \sum_{i=1}^{n_M} g(D_i).$$

If g is Lipschitz with constant L (a mild smoothness condition),

$$|g(D_i)| \leq L|D_i|.$$

Hence,

$$|\text{Bias}(\hat{\tau}^{\mathcal{M}} \mid X)| \leq L \cdot \frac{1}{n_M} \sum_{i=1}^{n_M} |D_i|.$$

Connecting to Chebyshev. Using the decomposition

$$|D_i| \leq |\mu_D| + |D_i - \mu_D|,$$

and applying Chebyshev to the second term, we obtain:

$$\frac{1}{n_M} \sum_{i=1}^{n_M} |D_i| \leq |\mu_D| + t + \frac{s_D^2}{t}.$$

Optimizing over t yields:

$$\frac{1}{n_M} \sum_{i=1}^{n_M} |D_i| \leq |\mu_D| + 2s_D.$$

Thus,

$$|\text{Bias}(\hat{\tau}^{\mathcal{M}} \mid X)| \leq L \left(|\Delta_X^{\mathcal{M}}| + 2s_D\right).$$

Implication. Matching reduces bias in treatment effect estimation by:

- shrinking the mean difference $|\Delta_X^{\mathcal{M}}|$,
- shrinking the dispersion s_D of pairwise differences.

Chebyshev's inequality formalizes the idea that:

$$\textbf{Good covariate balance} \implies \textbf{small bias in matched estimators}.$$

This mirrors the Week 1 result that regression bias arises from imbalance in $g(X)$; matching reduces this imbalance by construction.

5.9 Summary

Matching provides a principled, design-based way to reduce bias in observational studies. Pairwise covariate differences are the building blocks of balance, and Chebyshev-type arguments show that small variance in these differences ensures that most pairs are well balanced. Under mild smoothness assumptions on g, this balance translates directly into small bias in treatment effect estimation.

- Covariate balance measures (SMDs, variance ratios, distributional metrics) quantify how well matching has approximated randomization.
- Pairwise differences D_i are the building blocks of balance in matched designs.
- Chebyshev bounds show that small variance in D_i implies that most pairs are well balanced, even without distributional assumptions.
- Under mild smoothness assumptions on g, small pairwise differences imply small bias in treatment effect estimation.

Matching therefore provides a principled, design-based way to reduce bias in observational studies, with Chebyshev-type arguments offering transparent, assumption-light guarantees.

6 Worked Examples

6.1 Example 1: Digital Marketing

A firm wants to estimate the effect of receiving a promotional email ($Z = 1$) on purchase within 7 days (Y). Assignment is not random: more engaged users are more likely to receive the email.

Covariates:

- past purchase frequency,
- browsing intensity,
- device type,
- demographic attributes.

Design:

1. Specify a distance metric on covariates (e.g., Mahalanobis distance).
2. Use optimal matching to construct pairs of treated and control users with similar covariates.
3. Check balance on engagement metrics and demographics.

After matching, treated and control users have similar engagement histories, approximating a randomized experiment in the matched sample.

6.2 Example 2: Health Sciences

Goal: estimate the effect of a new diabetes management app on HbA1c levels.

Confounding arises because more health-conscious patients are more likely to adopt the app.

Design:

- Use optimal matching on baseline HbA1c, age, BMI, comorbidities, and prior utilization.
- Optionally impose calipers on key covariates (e.g., baseline HbA1c).
- Assess balance using standardized differences and variance ratios.

Matched comparisons yield a more credible estimate of the app's effect, with covariates used to refine the design rather than to rescue it ex post.

6.3 R Exercises: Matching Designs Using `approxmatch`

The `approxmatch` package is not merely a matching routine; it is a *design constructor*. It allows the investigator to specify:

- the number of units from each group in each stratum (e.g. 1:1, 1:k, or multi-group),
- exact matching constraints,
- near-fine balance constraints,
- distance structures combining Mahalanobis, propensity, or user-defined distances.

This section introduces students to these design choices through a sequence of exercises.

We assume a data frame `df` with:

- `Z`: treatment indicator (0/1),
- `Y`: outcome,
- `X1, X2, X3, ...`: covariates.

We create a group label compatible with `approxmatch`:

$$\texttt{df\$group} = \begin{cases} \text{treated}, & Z = 1, \\ \text{control}, & Z = 0. \end{cases}$$

6.3.1 Exercise 1: Building a Distance Structure

The function `multigrp_dist_struc` constructs a distance structure between groups. This structure is the backbone of all matching designs.

```
library(optmatch)      # must be loaded manually
library(approxmatch)

df$group <- ifelse(df$Z == 1, "treated", "control")

# Covariates for distance
vars <- c("X1", "X2", "X3")
```

```
components <- list(mahal = vars)
wgts <- 1

dist_str <- multigrp_dist_struc(.data      = df,
                                grouplabel = df$group,
                                components = components,
                                wgts       = wgts)
```

Design insight. The distance structure determines which units are "close" in covariates. Changing `components` or `wgts` changes the geometry of the design.

6.3.2 Exercise 2: 1:1 Matching (Basic Design)

We now construct a simple 1:1 matched design: one treated unit matched to one control.

```
design     <- c(1, 1)         # 1 treated, 1 control
indexgroup <- "treated"

res <- kwaymatching(distmat    = dist_str,
                    grouplabel = df$group,
                    design     = design,
                    indexgroup = indexgroup,
                    .data      = df)

matches <- res$matches
head(matches)
```

Design insight. This is the simplest matched-pair design. Each row of `matches` is a stratum with one treated and one control unit.

6.3.3 Exercise 3: 1:k Matching (Variable Ratio Design)

We now match each treated unit to *two* controls (1:2 matching). This increases precision when controls are plentiful.

```
design     <- c(1, 2)         # 1 treated, 2 controls
indexgroup <- "treated"

res_12 <- kwaymatching(distmat    = dist_str,
                       grouplabel = df$group,
                       design     = design,
                       indexgroup = indexgroup,
                       .data      = df)

matches_12 <- res_12$matches
head(matches_12)
```

Design insight. 1:k matching reduces variance of the estimator while preserving covariate balance. Students can compare balance and sample size across 1:1 and 1:2 designs.

6.3.4 Exercise 4: Exact Matching on a Key Covariate

Suppose `device` is a categorical covariate (e.g. mobile vs. desktop) that we want to match *exactly*.

```
design     <- c(1, 1)
indexgroup <- "treated"

res_exact <- kwaymatching(distmat     = dist_str,
                          grouplabel  = df$group,
                          design      = design,
                          indexgroup  = indexgroup,
                          .data       = df,
                          exactmatchon = "device")

matches_exact <- res_exact$matches
```

Design insight. Exact matching enforces perfect balance on a nominal covariate. Students can inspect strata to verify that device types match exactly.

6.3.5 Exercise 5: Near-Fine Balance on a Categorical Covariate

Suppose `race` is a categorical covariate. We want the *marginal distribution* of race to be similar across treated and control groups, but we do not require exact matching within each stratum.

```
design     <- c(1, 1)
indexgroup <- "treated"

res_fine <- kwaymatching(distmat        = dist_str,
                         grouplabel     = df$group,
                         design         = design,
                         indexgroup     = indexgroup,
                         .data          = df,
                         finebalanceVars = "race")

matches_fine <- res_fine$matches
```

Design insight. Near-fine balance ensures that the *overall* distribution of race is balanced, even if individual strata are not perfectly matched on race. This is a powerful design tool when exact matching is too restrictive.

6.3.6 Exercise 6: Covariate Balance Diagnostics

We use `covbalance` to evaluate balance before and after matching.

```
vars <- c("X1", "X2", "X3")

details <- c('std_diff', 'mean', 'function(x) diff(range(x))')
names(details) <- c('std_diff', 'mean', 'range')

cb <- covbalance(.data      = df,
```

```
                 grouplabel = df$group,
                 matches    = matches_fine,
                 vars       = vars,
                 details    = details)

cb$std_diff
```

Design insight. Students can compare:

- 1:1 vs. 1:2 balance,
- exact vs. near-fine balance,
- how distance structure affects balance.

6.3.7 Exercise 7: Treatment Effect Estimation in Different Designs

After constructing a matched design, we estimate the treatment effect using the matched sample.

```
matched_ids <- as.vector(matches_fine)
matched_df  <- df[matched_ids, ]

# Simple difference in means
with(matched_df, mean(Y[Z == 1]) - mean(Y[Z == 0]))
```

Design insight. Students can compare treatment effect estimates across:

- 1:1 matching,
- 1:2 matching,
- exact matching,
- near-fine balance designs.

This reinforces the idea that *design choices affect both balance and precision.*

6.3.8 Conceptual Link to Chebyshev and Bias

Across all designs:

- Matching reduces pairwise covariate differences D_i.
- Chebyshev bounds imply that small variance in D_i ensures most pairs are well balanced.
- Under mild smoothness assumptions on $g(X)$, small D_i implies small bias in $\hat{\tau}^{\mathcal{M}}$.

Thus, the `approxmatch` toolkit operationalizes the design-based principle:

$$\text{Better balance} \Rightarrow \text{smaller bias and more credible causal estimates.}$$

Conceptual Takeaways

- Observational studies require careful design because treatment is not randomized.
- Matching aims to reconstruct, in the observed covariates, the balance that randomization would have created.
- Matching is a design tool, not an estimator; analysis is performed on the matched sample.
- Approximate pairwise balance in covariates implies approximate balance in means and in linear combinations of covariates; Chebyshev-type bounds formalize this intuition.
- Covariates are used to refine the design and improve precision, echoing the role of covariates in randomized experiments.

Looking Ahead

Next week (Week 6), we formalize ignorability, propensity scores, and covariate adjustment, connecting the design principles of matching to identification assumptions in observational studies and to the regression-based balancing perspective developed in Week 1.